MIL-C-16310C
13 December 1967
SUPERSEDING
MIL-C-16310B
15 July 1954
(See 6.6)

MILITARY SPECIFICATION

CYLINDER COMPRESSED GAS

(COMPRESSED AIR AND CARBON DIOXIDE,

NONSHATTERABLE AND NONMAGNETIC)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers special high-pressure cylinders for storage of compressed gases.

1.2 Sizes. The cylinders shall be of the following sizes, as specified (see 6.2):

Size A - 15-pound capacity CO_2.
Size B - 50-pound capacity CO_2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY
MIL-P-15011 - Pallets, Material Handling, Hardwood, Post Construction, 4-Way.
MIL-P-15149 - Paint, Stencil.
MIL-P-15328 - Primer, Pretreatment (Formula No. 117 for Metals).
MIL-I-17214 - Indicator, Permeability, Low-Mu (Go-No Go).
MIL-V-17360 - Valves, Cylinder, Gas, Carbon Dioxide Fire Extinguishers.
MIL-C-17376 - Cap, Compressed-Gas Cylinder.

STANDARDS

MILITARY
MIL-STD-101 - Color Code for Pipeline and For Compressed-Gas Cylinders.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-278 - Fabrication, Welding and Inspection of Machinery, Piping and Pressure Vessels for Ships of the United States Navy.

DRAWING

MILITARY
810-1385867 - Decalcomania for Navy Gas Cylinders.

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:35Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

INTERSTATE COMMERCE COMMISSION
ICC3AA - Code of Federal Regulations - Interstate Commerce Commission Regulations for Transportation of Explosives and other dangerous articles by land and water in rail freight service and by motor vehicles (highway and water) (including specifications for shipping containers).

(Application for copies should be addressed to the Superintendent of Documents Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Material. Cylinders shall be made of corrosion-resistant steel in accordance with any of the analyses shown in table I. Cylinders may be made from pierced billets, cupped plates, or seamless tubing with either welded pressure heads or spun bottoms. When welded cylinders are supplied, the pressure heads used with tubes from a particular heat shall also be manufactured from a single heat in order simplify gunfire testing.

Table I. Chemical composition.

Carbon (max.)	Manganese (max.)	Phosphorus (max.)	Sulfur (max.)	Silicon (max.)	Copper (max.)	Nickel (min.)	Chromium (min.)	Titanium
Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	
0.12	2.0	0.04	0.04	1.0	0.5	8.0	17.0	1/
.08	2.0	.04	.04	1.0	.5	8.0	17.0	

1/ Titanium content shall be not less than 5 times the carbon content.

3.2 When welded cylinders are supplied, the joint design, heat treatment and welding procedure shall be in accordance with MIL-STD-278. Details shall be approved by the command or agency concerned before fabrication (see 6.5). Cylinders shall not be wirewound.

3.3 General. Cylinders shall be clean and free from scale and weld spatter both inside and out. They shall be of true form and of such shape that they will stand in an upright position. Cylinders shall be nonshatterable and have an average magnetic permeability per unit volume of not more than 2.0, when complete with valve, but without cap. They shall be equal to or better in stength to corresponding cylinders approved under Interstate Commerce Commission Regulations.

3.3.1 Spun bottom cylinders. The sound metal in the bottom of spun cylinders free from nometallic inclusions shall be not less than one and one-half times the minimum required cylinder wall thickness. Cylinder bottoms may be drilled at the point of closure and plugged.

3.3.1.1 Spun bottom cylinders shall show no leakage when tested with dry air as specified in 4.4.4.

3.4 Pressure rating. The cylinders shall be designed for a working pressure of 2,000 pounds per square inch gage (p.s.i.g.) and at test pressure of 3340 p.s.i.g. The wall thickness shall be such as to keep the maximum wall stress (p.s.i.) at the test pressure, to a value at least 10 percent below the 0.2 percent offset yield strength of the material.

3.5 Valve protection cap and flange. Size B cylinder shall be fitted with a neck ring or cap flange of austenitic corrosion-resisting steel securely held in place on the cylinder, and shall be provided with a valve protection cap conforming to MIL-C-17376 only when cylinders are fitted with valves.

3.6 When specified (see 6.2), cylinders shall be fitted with valves in accordance with MIL-V-17360, class b for size A cylinders and class c for size B cylinders.

3.7 Capacity. The minimum capacity of cylinders, at a nominal temperature of 70°F. shall be as shown in table II.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:35Z
Check the source to verify that this is the current version before use.

Table II. Capacity of cylinders.

Size	Capacity of cylinders	Minimum water volume
	Pounds CO_2	Cubic inches
A	15	615
B	50	2300

3.8 Size and weight. The size and weight shall be as shown in table III.

Table III. Size and weight.

Size	Outside diameter	Length 1/	Weight 2/
	Inches	Inches	Pounds
A	6-1/4 ± 1/2	28 ± 1	45
B	8-1/2 ± 1/8	56 ± 1	190

1/ Excluding valve protection cap.

2/ Approximate weight when empty.

3.9 Threading. Cylinders shall be tapped with 1-inch diameter National Gas taper pipe thread, and shall be in accordance with Handbook H28.

3.10 Marking.

3.10.1 Each cylinder shall be identified with figures and letters as follows (the serial numbers shall be in figures at least 1/4 inch high for size A cylinders and 1/2 inch for size B cylinders):

One side	Illustration
"Nonmagnetic," the pressure rating and "Spun" when spun cylinders are supplied.	Nonmagnetic 2,000 PSIG SPUN
Manufacturer's symbol-Manufacturer's serial number - letters "Nonshat"	MFG DO125 NONSHAT
Government inspector's stamp	XY
Other side	
Military Specification number and date (month and year)	MIL-C-16310 6/67
Original test data (month and year)	6/67

3.10.2 When cylinders are fitted with valves, the empty and fully charged weights, to the nearest 1/4 pound, shall be indented on the valve body in accordance with MIL-V-17360.

3.11 Painting and identification.

3.11.1 Primer. Cylinders, after being indented with markings specified in 3.10, shall be thoroughly cleaned and then given a coat of primer conforming to MIL-P-15328.

3.11.2 Identifying color. Cylinders shall be painted in accordance with MIL-STD-101 except that the letters "non-magnetic" shall be stencilled longitudinally above the letters "NONSHAT," using the same paint as used for the latter.

3.11.3 Stencilling. In addition to the painting specified in 3.11.1, each cylinder shall be stencilled with the name of the gas contained therein. The stencilling shall be lengthwise on the cylinder on diametrically opposite sides in letters at least 2 inches high. The letters "NONSHAT" and "NONMAGNETIC," at least 1 inch in height, shall be stencilled longitudinally on the cylinder, 90 degrees from the name of the gas. The stencilling paint shall be white conforming to MIL-P-15149.

Source: https://assist.dla.mil -- Downloaded: 202640-18T08:35Z
Check the source to verify that this is the current version before use.

3.11.4 Decalcomanias. Two decalcomanias bearing the name of the gas contained in the cylinder shall be affixed to the shoulder of the cylinder in the same place and between the markings specified in 3.10.1. There shall be no paint over the decalcomanias. Decalcomanias shall conform to Drawing 810-1385867.

3.12 Workmanship. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance.

4.2.1 Inspection lot. All cylinders of the same size presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.2.2 Sampling for examination. A random sample of cylinders shall be selected in accordance with table IV from each inspection lot of material for examination of visual, dimensional, and weight characteristics with lot acceptance based on the sampling inspection requirements in accordance with table IV.

Table IV. Sampling for visual, dimensional and weight examination
AQL (approx.) =1.5 percent defective.

Number of cylinders in an inspection lot	Number of cylinders in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 to 500	50	2	3
501 to 800	75	3	4
801 to 1300	110	4	5
1301 to 3200	150	5	6
3201 and over	225	8	9

4.2.3 Sampling for gunfire tests. Three cylinders representing each lot of steel entering into the manufacture of the cylinders on a contract or order shall be selected for the gunfire tests specified in 4.4.1. If the contract or order calls for less than 30 cylinders, only one sample cylinder is required.

4.3 Visual, dimensional, and weight examination. Each of the sample cylinders selected in accordance with table IV shall be visually and dimensionally examined and weighed to verify compliance with this specification. Any cylinder in the sample containing one or more visual or dimensional defects or over required weight shall be rejected, and if the number of defective cylinders in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Tests.

4.4.1 Gunfire test. Cylinders for gunfire test shall be selected in accordance with 4.2.3 and shall be shipped to the Commander, Naval Weapons Laboratory, Dahlgren, Va. 22448, marked "For test." The cylinder shall be fitted by the contractor with a satisfactory valve which can be opened and closed to check the pressure. The cylinders shall be charged with air at the Naval Weapons Laboratory, Dahlgren, Va., 22448, to a pressure of 2,000 p.s.i.g. While under this pressure, the cylinder shall be hit perpendicular to the longitudinal axis with a 40-mm armor piercing projectile at a velocity of 1,400 feet per second. Under this test, the cylinder wall shall remain in one piece, showing no fragmentation at any point of tear other than that generally encountered at the projectile entrance and exit of the cylinder. If any cylinder fails to pass this test, the contractor

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:35Z
Check the source to verify that this is the current version before use.

will be permitted to reheat-treat all the cylinders manufactured from the same heat of steel, and additional cylinders shall be selected as specified in 4.2.3. If the cylinders pass this retest all cylinders made from the same heat, provided they receive the same heat-treatment, will be accepted. Failure of any cylinder in the retest will cause rejection of all cylinders made from that heat.

4.4.2 If furnished with valve and charge, each cylinder selected in accordance with 4.2.2 shall be tested for leakage by submerging in water at a temperature of not less than 65°F. for a period of not less than 2 minutes. Any evidence of leakage shall be cause for rejection. When valves are not purchased with the cylinders, then this test shall be performed on all cylinders using a minimum number of suitable valves loaned by the manufacturer. If so desired, a test method utilizing a halide gas and a halide leak detector, and approved by the command or agency concerned, may be substituted for the above water submergence test.

4.4.3 Where spun bottom cylinders are supplied, a careful and frequent observation shall be made to assure that uniformity of the spinning and welding process is maintained. From each sample group of cylinders selected in accordance with table IV, another selection of 1 percent of the sample shall be made. The cylinders so selected shall be sectioned across the spun bottom and examined to determine conformance with 3.3.1.

4.4.4 Dry air tests. Prior to heat treatment the bottoms of spun cylinders shall be subjected to a dry air test pressure of not less than 2,000 p.s.i.g. for 1 minute with the inverted bottom filled with water.

4.4.5 Permeability test. Each cylinder selected in accordance with 4.2.2 shall be tested for magnetic properties. The magnetic permeability shall be determined by means of a permeability indicator of the go-no go type conforming with or equal to that specified in MIL-I-17214. The magnetic permeability shall not exceed 2.0 after completed fabrication.

4.4.6 Hydrostatic test. Each cylinder shall be hydrostatically tested as required by Specification ICC3AA of the Interstate Commerce Commission Regulations.

4.5 Inspection of preparation for delivery. The packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(Preparation for delivery requirements specified herein apply only to direct Government procurement. Preparation for delivery requirements between contractors and subcontractors shall be as specified in the individual order.)

5.1 Packing. Cylinders shall be packed for shipment in accordance with level A, B or C as specified (see 6.2).

5.1.1 Size A cylinders (see 6.2).

5.1.1.1 Level A. Size A cylinders shall be packed in boxes conforming to PPP-B-585, PPP-B-601, or PPP-B-621 for overseas shipment. The gross weight shall not exceed approximately 200 pounds.

5.1.1.2 Level B. Size A cylinders shall be packed in boxes conforming to PPP-B-585, PPP-B-601 or PPP-B-621, for domestic shipment. The gross weight shall not exceed approximately 200 pounds.

5.1.1.2.1 Level B alternative. As an alternate method, size A cylinders may be packed on pallets conforming to MIL-P-15011. The cylinders shall be secured to the pallet with coated or galvanized 1-1/4 by 0.035 inch flat steel strapping, or equivalent round steel wire. Additional wood bracing shall be provided as necessary to adequately secure the load. The gross weight shall not exceed approximately 2100 pounds.

5.1.1.3 Level C. The cylinders shall be prepared for shipment in a manner which will insure acceptance by common carrier and safe arrival at destination.

5.1.2 Size B cylinders. Unless otherwise specified in the contract or order, size B cylinders shall be packed in pallets conforming to MIL-P-15011. Loading, strapping and wood collar details shall be as shown on figure 1. The strapping shall be coated or galvanized. The gross weight shall not exceed approximately 2100 pounds.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:35Z
Check the source to verify that this is the current version before use.

5.2 Marking. In addition to any special marking required in the contract or order, marking for shipment shall be in accordance with MIL-STD-129. Nomenclature shall be the exact nomenclature for the part or item that has been approved under the contract or order.

6. NOTES

6.1 Intended use. The intended use for cylinders covered by this specification is as follows:

(a) Size A, 15-pound capacity cylinders - For use as portable fire extinguishers (see MIL-E-24269).
(b) Size B, 50-pound capacity cylinders - For use in hose reel or fixed, built in fire extinguishers systems (see MIL-E-2185 and MIL-E-2186).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of cylinder required (see 1.2).
(c) Whether valves are to be shipped with cylinders, and type of valve (see 3.6).
(d) For size A cylinders, level of packing (see 5.1 and 5.1.1).
(e) That for Naval purchases the cylinders for gunfire tests will be shipped to the Naval Weapons Laboratory, Dahlgren, Va. 22448, at the contractor's expense; that the gunfire test will be made at the Government's expense; and that if the contractor desires return of the cylinders, they will be returned at his expense.

6.3 Cylinders ordered in accordance with this specification are empty and, unless otherwise specified by the command or agency concerned, without valves.

6.4 For Naval purchases. The Department of the Navy will arrange to conduct a reasonable number of preliminary gunfire tests, entirely at the expense of the exhibitors, in order that prospective bidders may obtain data upon the characteristics necessary in the material of cylinders to enable them to pass the gunfire test. Such tests will be considered as for the information of the bidders only, and regardless of the outcome it will be necessary that the requirements specified in 4.2.3 and 4.4.1 be fulfilled upon the award of any contract.

6.5 When welded cylinders are procured by a Naval agency or for ultimate use by the Navy, approval in accordance with 3.2 should be obtained from the Naval Ship Engineering Center, Department of the Navy, Washington, D. C. 20360.

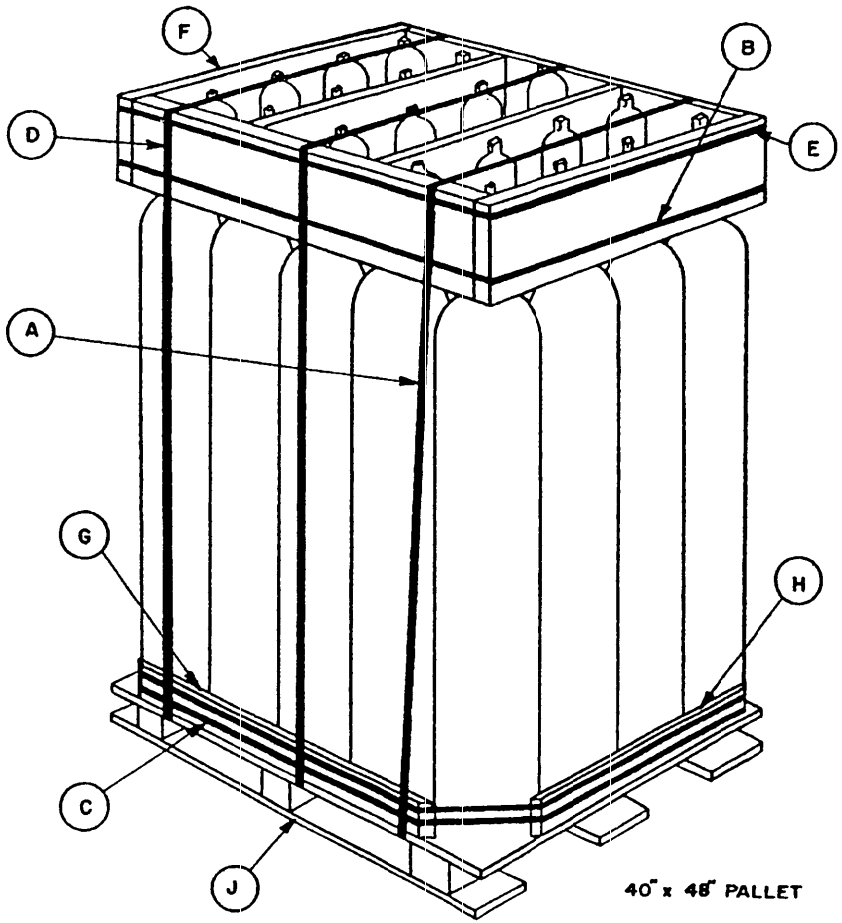
6.6 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Custodians:
Navy - SH
Air Force - 11

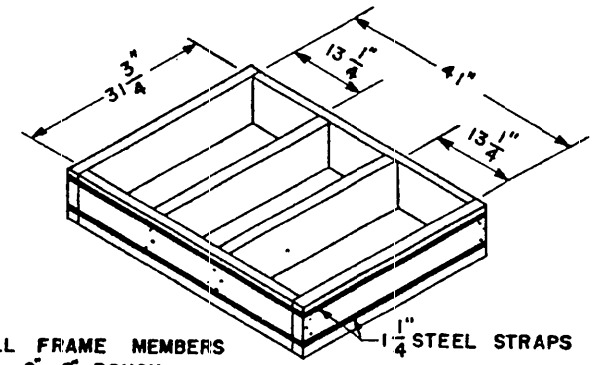
Preparing activity:
Navy - SH
(Project 8120-0272)

Review activities:
Navy - SH, YD
Air Force - 11, 68

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:35Z
Check the source to verify that this is the current version before use.



USE 5-12 d NAILS
AT EACH JOINT.



DETAIL OF WOOD COLLAR

BILL OF MATERIALS		
ITEM	NO REQ'D	DESCRIPTION
A	3	1 1/4" x .035" x 17'-0" STEEL STRAPS OR EQUIVALENT ROUND STEEL WIRE
B	2	1 1/4" x .035" x 14'-6" STEEL STRAPS OR EQUIVALENT ROUND STEEL WIRE
C	2	1 1/4" x .035" x 14'-6" STEEL STRAPS OR EQUIVALENT ROUND STEEL WIRE
D	7	1 1/4" STRAPPING SEALS
E	24	1 1/4" STAPLES
F	1	WOOD COLLAR (SEE DETAIL)
G	2	2" x 4" x 37" BATTEN, BEVELED
H	2	2" x 4" x 29" BATTEN, BEVELED
J	1	40" x 48" PALLET
	40	12 d COATED NAILS

FIGURE 1. *Unit load for cylinders.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:35Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:35Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE
(NAVSHIPS OVPRT 12-66)

Source: https://assist.dla.mil -- Downloaded: 202640-18T08:35Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL SHIP ENGINEERING CENTER
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER, NAVAL SHIP ENGINEERING CENTER
FEDERAL & MILITARY DOCUMENTS & DOD STANDARDIZATION BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:35Z
Check the source to verify that this is the current version before use.